

Database Connection Troubleshooting

Error: “db_config_local.m not found”

Solution: 1. Copy the template: `copyfile('config/local/db_config_local.m.template', 'config/local/db_config_local.m')` and fill in your credentials (Username, Password). It's gitignored — never committed. 2. Make sure config is on your MATLAB path (run `startup.m`)

Error: “Failed to establish database connection”

Check 1: Verify Database Toolbox

```
license('test', 'database_toolbox')
```

Should return 1. If not, install Database Toolbox.

Check 2: Test basic MATLAB database function

```
% For SQL Server
conn = database('NARWCDB', 'username', 'password', ...
    'Vendor', 'Microsoft SQL Server', ...
    'Server', 'your-server');
isopen(conn)
```

Check 3: Verify server connectivity

Can you ping the server?

Is the database server running?

Are you on the correct network/VPN?

Check firewall settings

Check 4: Verify credentials

Username correct?

Password correct (no typos)?

User has permission to access database?

Error: “Master table not found”

The database connection works, but the Master table doesn't exist yet.

Solution: You'll create this in Phase 1 (migration). For now, this is expected.

Error: “Connection timeout”

Possible causes:

Server is slow to respond

Network issues

Server is overloaded

Solution: Increase the timeout—either in `config/defaults/db_config_default.m` (project-wide default) or `config/local/db_config_local.m` (per-machine override):

```
db.Timeout = 30; % seconds
```

SQL Server Specific Issues ODBC Driver Issues

If using SQL Server, you may need to install/configure ODBC drivers.

Check available drivers:

```
drivers = listdrivers
```

Install ODBC Driver:

Windows: Download "ODBC Driver for SQL Server" from Microsoft

Mac/Linux: Install unixODBC and FreeTDS

Windows Authentication

For Windows authentication, use:

```
db.Username = '';  
db.Password = '';
```

MySQL Specific Issues Can't connect to MySQL server

Verify MySQL is running: `sudo service mysql status`

Check port is correct (usually 3306)

Verify user has remote access permission

Access denied error

-- In MySQL, grant permissions:

```
GRANT ALL PRIVILEGES ON NARWCDB.* TO 'username'@'%' IDENTIFIED BY 'password';  
FLUSH PRIVILEGES;
```

PostgreSQL Specific Issues Connection refused

Check PostgreSQL is running

Check `pg_hba.conf` allows your connection

Verify port (usually 5432)

SSL required error

Add to connection string or configure PostgreSQL to not require SSL.

Quick Start Commands

```
% === Complete test sequence ===

% 1. Setup project
startup

% 2. Quick connection test
test_connection

% 3. If connection works, test with a query
conn = narwc.db.Connection.create();
data = conn.fetch('SELECT TOP 10 * FROM Master');
disp(data)
conn.close()

% 4. Run unit tests (more thorough)
runtests('tests/unit/test_db_connection.m')
```

Expected Output (Success)

```
=== Database Connection Test ===

Test 1: Loading configuration...
  ✓ Configuration loaded
    Type: SQLServer
    Server: your-server-name
    Database: NARWCDB

Test 2: Creating connection object...
  ✓ Connection object created

Test 3: Checking connection status...
  ✓ Connection is open

Test 4: Executing test query...
  ✓ Query executed successfully
    Database version: Microsoft SQL Server 2019...

Test 5: Checking for Master table...
  ✓ Master table exists
    Record count: 12450

Test 6: Testing sample data query...
  ✓ Sample query successful
    Retrieved 5 records
```

Sample data:

FILEID	YEAR	EVENTNO
'F098027'	1998	33
'F098027'	1998	34
...		

Test 7: Closing connection...

✓ Connection closed successfully

=== Connection Test Complete ===

If all tests passed, your database connection is working!